

AI practical on highly constrained embedded devices.

Q. Any measurable performance gains compared to other runtimes?

A. Direct performance comparisons vary depending on the model and workload. However, Alif's microcontrollers demonstrate more than a 100x speed-up compared with traditional microcontrollers without NPUs. Full hardware acceleration delivers strong performance per watt, improving efficiency for AI inference on constrained edge devices.

Q. How is generative AI reshaping embedded applications?

A. Generative AI on embedded hardware enables new use cases, including real-time speech-to-text

processing, conversational toys, contextual personal assistants, and interactive voice systems. Because transformer models incorporate a form of memory, they can maintain context across multiple interactions, enabling multi-turn conversations entirely on-device.

Q. What challenges arise when running generative models fully on-device?

A. Running generative models fully on-device introduces challenges such as including strict power limits, thermal constraints, and limits on locally stored conversational context. Despite these hurdles, the benefits include near-zero latency and improved privacy, as user data remains on the device.

Q. How does Alif achieve high AI performance while keeping power low?

A. Alif achieves this balance through a split-silicon architecture that separates responsibilities across two domains. A high-efficiency domain manages always-on sensing and low-power machine-learning triggers, handling continuous data collection. A high-performance domain activates only when more complex analysis is required, delivering rapid processing without unnecessary power draw. **END** 

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/EFYInt3feb26>



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Users simply provide their datasets, select the forecasting model and generate predictions without rebuilding the entire pipeline. Outputs are exported as APIs and JSON files, allowing seamless integration with municipal command centres and government health portals.

The system has already been integrated with Bengaluru's BBMP command and control centre.

Although dengue served as the pilot use case, the platform has also been designed for broader public health applications. The same infrastructure can be adapted for forecasting malaria, heat stress and other climate-sensitive diseases by replacing datasets and models while retaining the underlying architecture.

ARTPARK is also working to incorporate additional datasets such as mosquito vector density, mobility

patterns and serotype information to improve forecasting performance.

Current progress and future roadmap

The platform currently delivers up to four-week forecasts with approximately 80% accuracy at district and sub-district levels. The team has also made ward-level predictions operational and is expanding deployments beyond Karnataka to states like Odisha and Andhra Pradesh.

An open data catalogue has been released, allowing researchers to access standardised datasets and documentation for developing and benchmarking their

own forecasting models.

Future plans include integrating more AI models, enhancing visualisation tools, improving API integrations with national health platforms, and opening the ForecastHub to external contributors.

By combining open data, modular AI models and reusable software infrastructure, ARTPARK aims to create a collaborative ecosystem where disease forecasting becomes a shared public capability rather than isolated research, helping governments move from reactive outbreak management to proactive public health preparedness. **END** 

This article is based on the talk 'Open Source Infrastructure for Scalable, Reproducible Disease Forecasting' by Dr Raunaq Pradhan, product manager, ARTPARK, Indian Institute of Science, Bengaluru, at Open Source India 2025. It has been transcribed by Nikita Kumari, journalist at the Efy Group.